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A study on the cell life cycle: A random sample of cells was taken to measure the effect of external factors such as temperature and carbohydrate consumption rate on the cell life cycle, and to study the differences between these factors.

```
##
## 1 function (... , list = character(), package = NULL, lib.loc = NULL,
## 2     verbose = getOption("verbose"), envir = .GlobalEnv, overwrite = TRUE)
## 3 {
## 4     fileExt <- function(x) {
## 5         db <- grepl("\\\\.([^.]+)\\. (gz|bz2|xz)$", x)
## 6         ans <- sub(".*\\\\\\. ", "", x)
```

Table one : shows the number of study variables and the number of individuals.

```
##      V1  V2 V3
## 1  150 0.6 10
## 2   86 1.0 10
## 3   49 1.4 10
## 4  288 0.6 20
## 5  157 1.0 20
## 6  131 1.0 20
## 7  184 1.0 20
## 8  109 1.4 20
## 9  279 0.6 30
## 10 235 1.0 30
## 11 224 1.4 30
```

```
## [1] 11  3
```

```
## 'data.frame':   11 obs. of  3 variables:
## $ V1: int  150 86 49 288 157 131 184 109 279 235 ...
## $ V2: num  0.6 1 1.4 0.6 1 1 1 1.4 0.6 1 ...
## $ V3: int  10 10 10 20 20 20 20 20 30 30 ...
```

Table 1: The Descriptive Statistic for Data

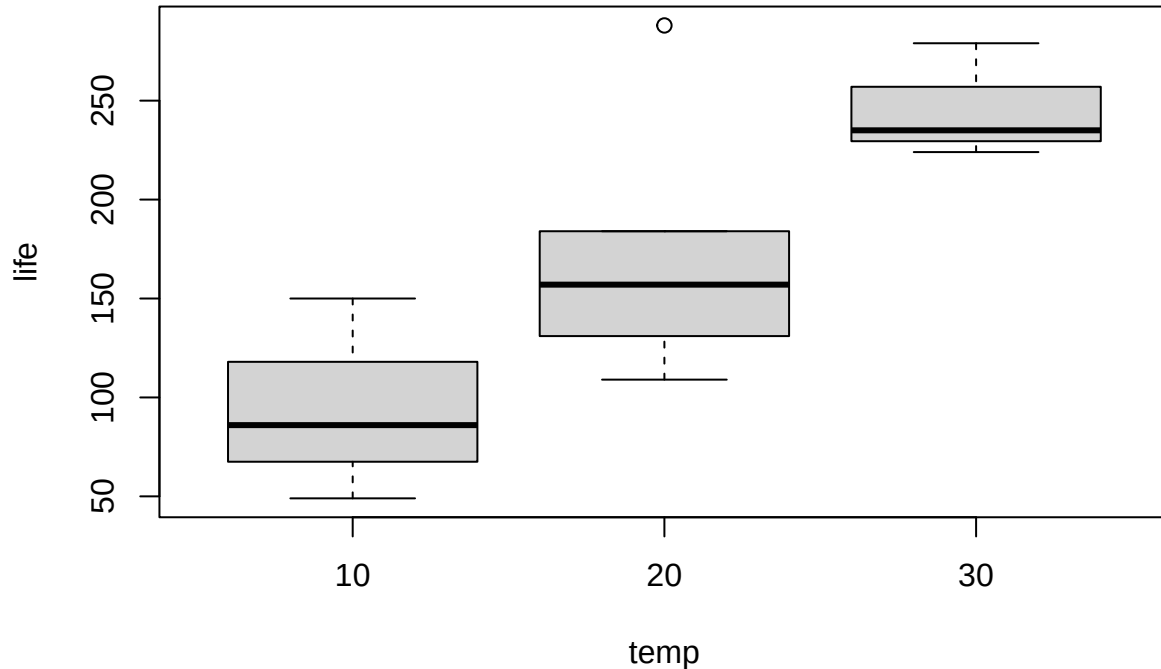
life	chrates	temp
Min. : 49.0	Min. :0.6	10:3
1st Qu.:120.0	1st Qu.:0.8	20:5

life	chrates	temp
Median :157.0	Median :1.0	30:3
Mean :172.0	Mean :1.0	NA
3rd Qu.:229.5	3rd Qu.:1.2	NA
Max. :288.0	Max. :1.4	NA

Table two: shows that the mean value of the variable (life) is equal to 172 and And greater value is 288 and Smallest value is 49 , The results show a significant deviation in the values, indicating a lack of homogeneity, In terms of chrates carbohydrate intake, it appears that carbohydrate consumption values are homogeneous, with the highest value reaching [a certain level] is 1.4 and lowest levels is 0.6 , The number of cells at 10 degrees Celsius was 3, the number of cells at 20 degrees Celsius was 5, and the number of cells at 30 degrees Celsius was 3.

```
## [1] 11 3
```

```
## 'data.frame': 11 obs. of 3 variables:
## $ life : int 150 86 49 288 157 131 184 109 279 235 ...
## $ chrates: num 0.6 1 1.4 0.6 1 1 1 1.4 0.6 1 ...
## $ temp : Factor w/ 3 levels "10","20","30": 1 1 1 2 2 2 2 2 3 3 ...
```



The graph shows that the average cell lifespan at 30 degrees is greater than at 10 and 20 degrees.

Inference statistic:

```
## ANOVA Table (type II tests)
```

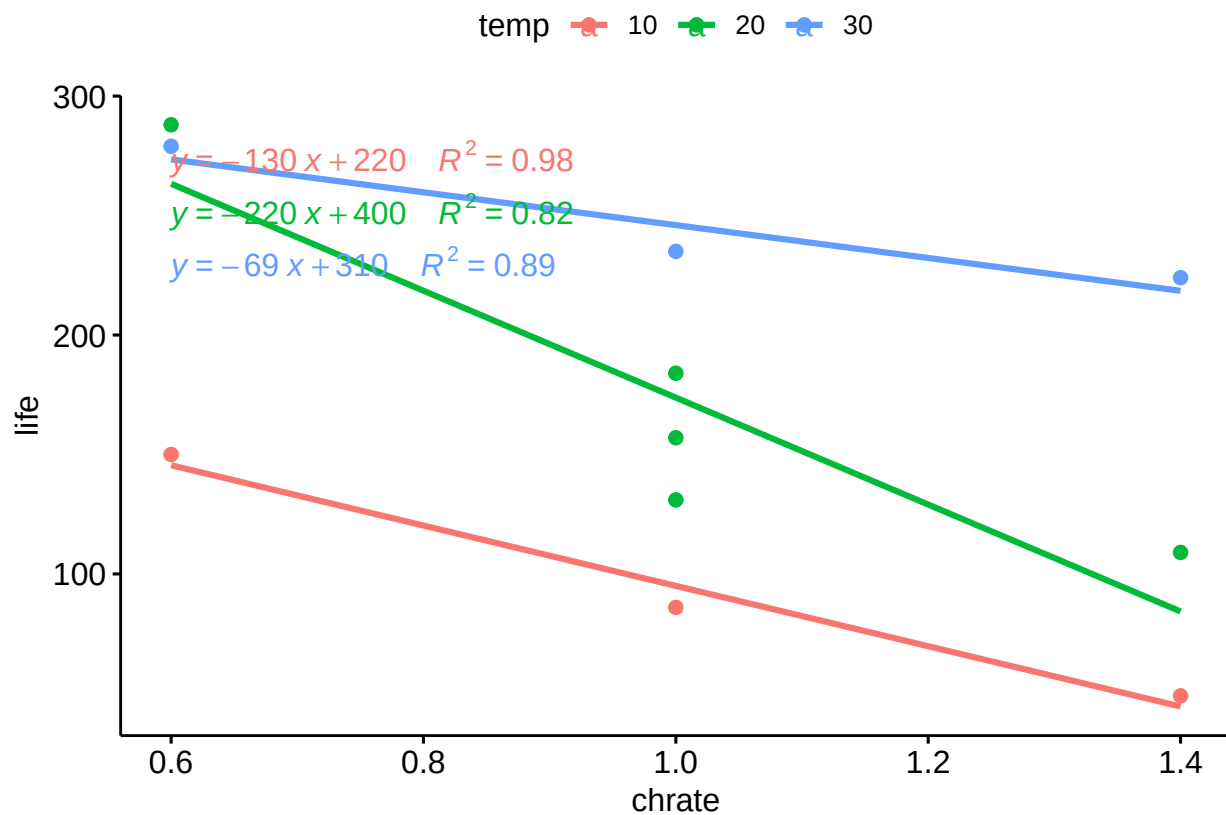
```
##
##   Effect DFn DFd      F      p p<.05   ges
## 1 chrates 1    7 17.069 0.004    * 0.709
## 2 temp    2    7 15.619 0.003    * 0.817
```

Table 2: ANOVA Table

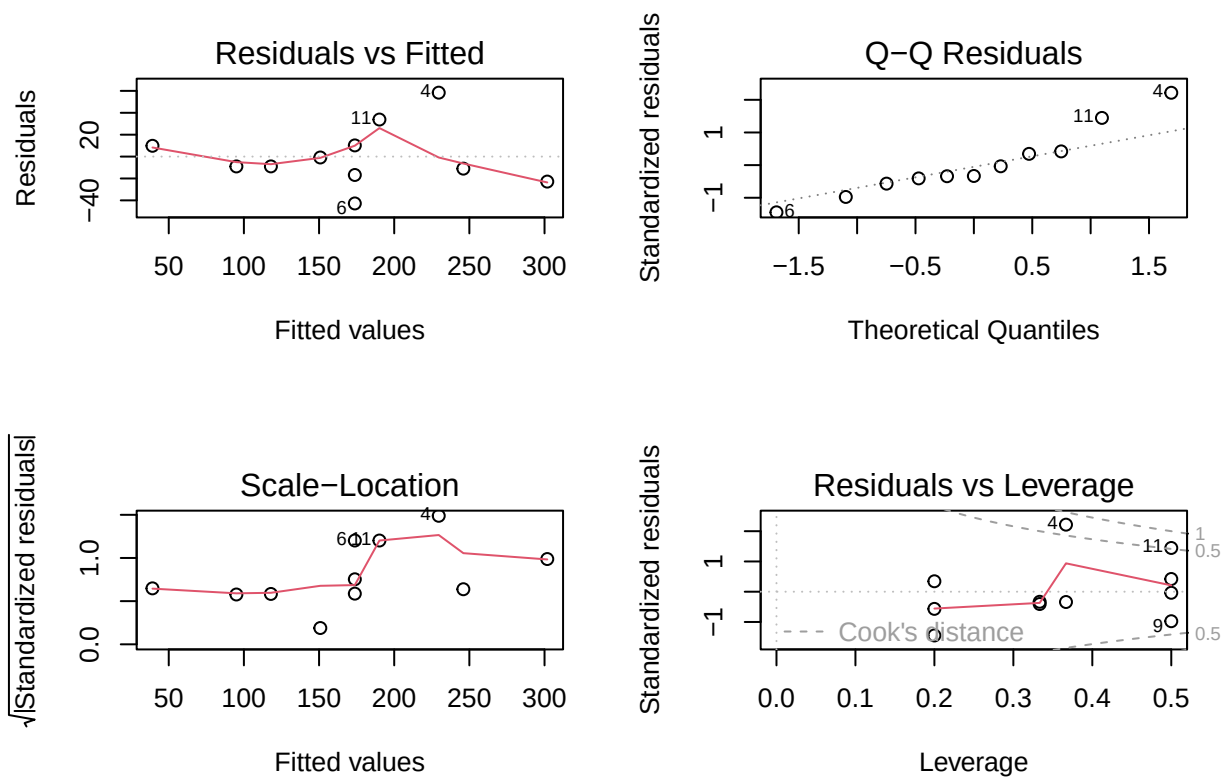
Effect	DFn	DFd	F	p	p<.05	ges
chrates	1	7	17.069	0.004	*	0.709
temp	2	7	15.619	0.003	*	0.817

The results showed that there were statistically significant differences amounting to a value p-value of chrates is 0.004 less than This suggests that carbohydrate levels had an effect on cell life It has an effect on modeling, The results showed that there were statistically significant differences between the average temperatures, where the value reached p-value is 0.003 less than 0.05

Check the assumption:



The diagram showed that there is a strong negative linear relationship between cell life and carbohydrate content at 10°C The carbohydrate content explained 0.98% of the variation in cell life, meaning it had a significant impact, At a temperature of 20, the model interprets its ratio is 0.82 This indicates that the rate of carbohydrate consumption explains 0.82% of cell life, At a temperature of 30 degrees, the rate of carbohydrate consumption was interpreted as follows is 0.89.



Residuals vs fitted plot showed random scatter, indicating linearity and constant variance assumptions were reasonably met, The Normal Q-Q plot showed points far away the line, not normal distribution Scale-location not homogeneous because the points are not distributed horizontally. Residuals vs leverage plot showed that the points are inside shape, no influential points.